

Intermediate Presentation

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Planned Algorithms

Baseline: Recommendations from the TMDB API filtered to match the MovieLens Dataset

1. Collaborative Filtering (Item-Item kNN)

... finds the highest rated movies from users that also gave the reference movie a high rating (sklearn)

2. Content-Based Filtering (Actor Overlap)

... identifies movies that have the highest actor, director, or crew overlap with the reference movie (user's preferences)

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3. Collaborative Filtering (Genome Scores)

... using the genome tags and the genome scores to recommend movies with similar semantic profiles

4. Visual Mood (Image Based – Backdrops)

... identifies the movies with the highest visual similarity based on the backdrops (TMDB) an a pre-trained CNN (ResNet, ...)

5. Subtitle Recommender (Deep Learning)

... using the subtitles of the movie (TF-IDF, Sentence Transformer) to recommend movies with similar tones.

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6. Hybrid Recommender (Weighted RRF)

... identifies the best movie by sorting them according to the ranks across the other algorithms, using the user's preferences as weights.

7. Collections (Bonus)

... also listing movies from the same collection